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Module 03 Lab: Writing a Case-Swap Program Using Branching Logic

1. Using a 2-digit hexadecimal value, you can represent 256 ASCII characters as 16 x 16= 256.

   Other character encoding standards include Unicode and ISO. Unicode has different lengths: UTF-8, 16, 32, and is found in JavaScript and Windows. ISO, similar to ASCII with region-specific characters, is another standard found in Western European languages.

   With letters, uppercase and lowercase are separated by 32 bits. Meaning to convert from lower to upper, one can add 32 and subtract for vise versa.

2. uNc cHArLOtTe

   i lOvE risc-v

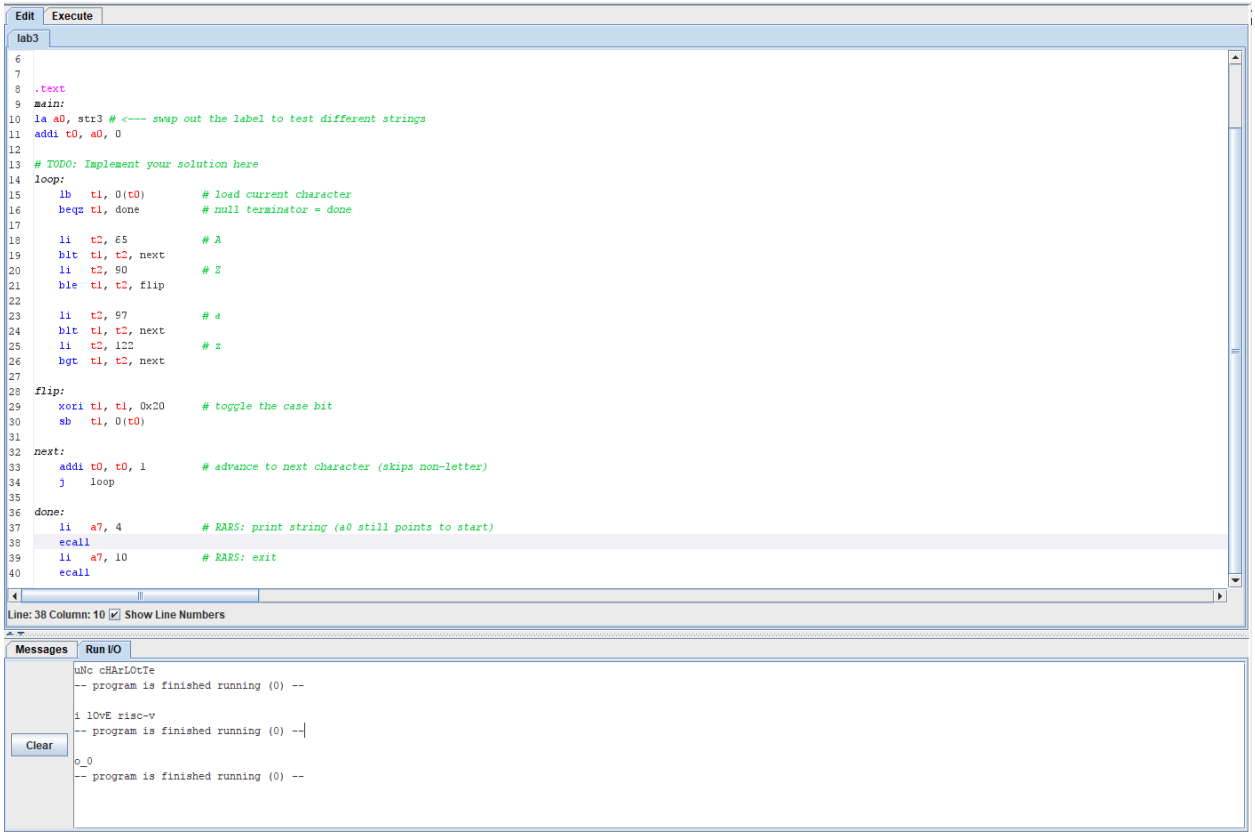
   o_0

3. Printing the string makes debugging easier, as you can visually check the final output to see if there were any errors.

   Two other things environmental calls can do are exit the RARS program and read from user input.

4. A FOR loop would be the most appropriate, as we need to scan each bit individually to see if it's a case-swappable character.

   We can identify an ASCII letter by its decimal value. 65-90 for A-Z and 97-122 for a-z. All other decimal values refer to other characters.

5. 

Yes, I have some background in coding, and this follows standard looping procedures.

No, I believe the difficulty would be the same. If implemented, only the range and condition values would change; the logic would still function the same.

Using a Java loop (mimic logic):
String a0 = “"UnC ChaRloTtE";
String low = a0.charAt(0).toLowerCase();
String up = a0.charAt(0).toUpperCase();
For (int i = 0; i < a0.length(); i++) {
If (a0.charAt(i).equals(low)) {
a0.charAt(i) = a0.charAt(i).toUpperCase();
} else If (a0.charAt(i).equals(up)) {
a0.charAt(i) = a0.charAt(i).toLowerCase();
} else {
// nothing happens, non-letter
}
}
System.out.println(a0);
Note this does not actually work due to char/string class variable differences.

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The ASCII version is definitely longer than Java. Being assembly code, ASCII lacks the complex shortcuts and multivariable store/call functions. So while sharing similar logic, ASCII generally tends to be longer.